

Effects of Observing a Robot Expressing Approach-Avoidance Conflict on Observers' Self-Evaluations of Personal Courage

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Abstract

Robots can externalize pre-action internal states in a controlled and reproducible form, but it remains unclear whether observing such states influences how people evaluate their own capacity to act despite fear. We examined whether a robot's visual externalization of approach-avoidance conflict before action influences observers' self-evaluations of personal courage. Speech bubbles were projected near the head of a cylindrical robot to present approach and avoidance motives in a scenario in which the robot encountered a person littering in a park. Study 1 compared conflict and no-conflict conditions and sequential and simultaneous presentation methods to determine whether a robot expressing approach-avoidance conflict was perceived as courageous. Study 2 varied whether the robot expressed one motive direction or both approach and avoidance motives and whether it admonished the litterer, and examined whether post-stimulus courage self-evaluations differed according to observers' preexisting courage tendency. In Study 1, the conflict condition yielded higher courage and conflict ratings than the no-conflict condition. In Study 2, the predicted conflict-by-action pattern was not supported; instead, conflict interacted with preexisting courage tendency. The low-courage group showed a marginal trend toward higher self-evaluations in the conflict condition, whereas participants in the high-courage group reported lower self-evaluations in the conflict condition. Admonishing behavior had no clear effect. These findings indicate that robot-expressed motivational conflict is associated with different self-evaluation patterns depending on observers' prior characteristics. Methodologically, the paradigm illustrates how robots can serve as controlled social models for externalizing pre-action internal states that are difficult to observe or standardize in human models.

Introduction

In everyday life, people sometimes must pursue valued actions even when those actions involve fear or anxiety. In this study, we refer to this capacity as personal courage. Courage has been conceptualized not as the absence of fear but as moving toward meaningful action in the presence of fear (Rachman, 1984; Woodard and Pury, 2007). Personal courage is important because it situates the difficulty of an action within the actor's own context. Pury et al. (2007) distinguished between general courage, which is evaluated as courageous by most people, and personal courage, which is courageous in light of a person's life context and personal constraints. From this perspective, a child with a learning disability going to school on the day of a major test, or a person wrapping Christmas presents for the first time after being unable to do so for many years because of a history of post-traumatic stress disorder, can be understood as performing actions that involve high fear and difficulty within that person's context (Pury et al., 2007). Furthermore, actions such as speaking up, seeking help, or pointing out a problem may be valuable for the person or for others, but they also involve social and psychological risks, such as negative evaluation and rejection. These actions are therefore meaningful targets for investigation from the perspective of personal courage (Milliken et al., 2003; Vogel et al., 2007; Howard et al., 2017).

However, people do not always act even when they recognize that an action is valuable. For example, expressing an opinion in front of others often involves anxiety about how one will be evaluated (Watson and Friend, 1969; Dickerson and Kemeny, 2004). Consulting others or seeking help can also involve concern about being viewed negatively (Vogel et al., 2007). In workplaces, many employees report having noticed problems or concerns but not communicating them to supervisors, mainly because they fear being viewed negatively or damaging important relationships (Milliken et al., 2003). Thus, speaking up and seeking help can benefit oneself and others, yet they involve social and psychological risks, such as negative evaluation, damaged relationships, reduced status, and damage to self-image. Therefore, supporting personal courage requires more than helping people understand that an action is valuable; it also requires supporting the process of moving toward action despite perceived risk.

As one way to support this process, this study focuses on observational learning, in which observing another agent's behavior or success can change expectations about one's own ability to act (Bandura, 1977). Prior research has shown both changes in externally observable motor performance following model observation (Weiss et al., 1998) and motor-related neural activity during action observation (Chaisaen et al., 2020). Observational learning can also influence psychological processes such as self-efficacy and persistence (Schunk and Hanson, 1985; Leonard et al., 2017). In particular, observing a model struggle before succeeding may communicate not only what action is possible but also that difficulty can be overcome. This possibility is relevant to personal courage, for which fear and hesitation before action are central.

However, fear and hesitation are covert states that are difficult to present in a standardized form using human models. Robots can explicitly externalize such pre-action states while keeping their appearance and overt behavior relatively consistent across conditions, thereby providing a controlled social model for examining how expressed conflict relates to observers' self-evaluations. Accordingly, this study examined whether observing robot-expressed approach-avoidance conflict, defined as the coexistence of motives to pursue a valued action and to avoid its risks, influences observers' immediate self-evaluations of personal courage and whether this influence depends on their preexisting courage tendency.

We conducted two studies. Study 1 examined whether a robot expressing approach-avoidance conflict was perceived as more courageous than a robot expressing motives in a single direction and selected the presentation format for Study 2. Study 2 crossed motivational structure (univalent vs. conflicted) with behavioral outcome (admonition vs. no admonition) and examined whether the resulting self-evaluation patterns differed according to observers' preexisting courage tendency. Together, the studies tested whether

externalizing a robot's pre-action conflict changes how its behavior is perceived and whether observers' responses vary according to their prior characteristics. This question is relevant to the design of expressive social robots because displaying hesitation may not have uniform consequences across users.

Related Work

Personal Courage and Internal States Before Action

Research on courage has treated courage not as the absence of fear but as moving toward action while experiencing fear. Rachman (1984) argued that even people with strong fear can perform courageous acts. Norton and Weiss (2009) examined the relationship between behavioral approach and courage in a fear-eliciting situation. Woodard and Pury (2007) also conceptualized courage as a construct involving action for a meaningful purpose in a situation involving threat or fear. These studies provide a basis for understanding courage not as mere risk preference or low fear but as approach behavior in the presence of fear.

Within this literature, Pury et al. (2007) distinguished between general courage, which is evaluated as courageous by most people, and personal courage, which is courageous in light of a person's life context and personal constraints. Actions involving high personal courage are characterized by fear, difficulty, and personal constraints. According to this distinction, the same action may or may not be courageous depending on the fear or difficulty the actor experienced. Therefore, research on personal courage needs to consider not only externally visible behavior but also the internal states preceding that behavior.

Recent process models of courage also focus on these internal states before action. Chowkase et al. (2024) describe courage as a sequence involving situation perception, evaluation of value or meaning, evaluation of action feasibility, and action decision. In this process, reasons to move toward a valued action and reasons to avoid risks or disadvantages associated with the action may coexist. This coexistence can be understood as approach-avoidance conflict, in which approach and avoidance motives operate simultaneously (Lewin, 1931; Miller, 1944). However, these studies clarify the definition and process of courage; they do not directly examine whether observing another agent's internal state changes observers' self-evaluations of personal courage.

Observational Learning and Dependence on Observers' Prior Characteristics

Beyond a model's final behavior, the process displayed before an outcome may also shape observational learning.

Schunk et al. (1987) compared mastery models who succeeded easily from the beginning with coping models who showed difficulty and gradually coped with it, and found that observing coping models increased self-efficacy and task performance. Schunk and Hanson (1989) also showed that models who cope while displaying difficulty and negative affect influence learners' self-efficacy. Leonard et al. (2017) found that infants who observed an adult persist in attempting to achieve a goal made more attempts on a subsequent difficult task than infants who observed an adult succeed easily. However, the relative advantage of coping models over mastery models has not been consistently observed. Schunk and Hanson (1985) found that children who observed a peer model showed higher self-efficacy and achievement than those who observed a teacher model or no model, but reported no significant differences between the mastery- and coping-model conditions. Taken together, these findings suggest that observing a process involving difficulty and effort can support observers' self-efficacy and persistence, although coping models do not necessarily outperform mastery models.

At the same time, the same model may be interpreted differently depending on observers' prior characteristics. Braaksma et al. (2002) showed that similarity between model and observer and the model's level of expertise are important in observational learning. Lucas et al. (2006) showed that people tend to rely on others' judgments and behavior as cues when a task is difficult, whereas people with high self-efficacy are less susceptible to such social influence. These findings indicate that, when using observational learning, it is necessary to consider not only the model's characteristics but also observers' prior characteristics, which influence how the model is received. Therefore, the present study also considers observers' prior characteristics when examining the effect of a robot's conflict expression.

Modeling by Robots and Artificial Agents

Models in observational learning are not limited to humans. Studies using artificial agents and avatars have shown that visually present agents and self-similar avatars can influence observers' attitudes, self-efficacy, and subsequent behavior (Rosenberg-Kima et al., 2008; Fox and Bailenson, 2009). Research on social robots has also shown that people can treat robots as social models and engage in modeling based on their behavior and social roles (Xu, 2023). Prior research has also revealed that the way a robot encourages participants influences how participants subsequently encourage others (Higashino et al., 2023).

These studies indicate that robots and artificial agents can function as social models that influence observers. However, the studies reviewed here targeted attitudes, self-efficacy, movement, and encouragement behavior, not personal courage itself. Within the scope of the literature reviewed in this manuscript, few past reports have explicitly discussed the fear or conflict experienced by a model before action and examined how observing that state influences observers' self-evaluations of personal courage. Therefore, using a robot to present internal states related to personal courage in an observable form extends existing modeling research.

Externalizing Internal States Through Robots

The experimental rationale for using a robot in this study is that visual expressions, along with behavior and speech, can be controlled more easily across conditions. With a human model, observers can infer internal states from facial expressions, voice, silence, or self-disclosure. However, to present the presence or absence of fear, hesitation, approach motives, and avoidance motives in the same scenario and in a comparable form across conditions, an artificial medium is more suitable. Therefore, in this study, we used speech-bubble expressions in addition to the robot's movement and speech, and presented the robot as a stimulus in which internal states were externalized.

Regarding this display format, Nitada et al. (2021) proposed a method in which a speech bubble is displayed near a robot's head in a video-conferencing situation, with the user's face drawn inside the bubble. They found that the speech-bubble condition increased the sense that the robot was looking at the user. Their study suggests that speech bubbles can function as a means of visually representing the robot's attention or mental focus. However, that study addressed the direction of attention, not the conflict between approach and avoidance motives related to courage.

Taken together, prior studies have shown that personal courage involves actions accompanied by fear or difficulty, that observational learning can influence self-efficacy and perceived action possibility, and that robots and artificial agents can function as social models. However, within the scope of the literature reviewed in this manuscript, research has not sufficiently examined whether externalizing internal states before action related to personal courage through a robot, especially conflict between approach and avoidance motives, affects observers' self-evaluations of personal courage. The present study focuses on this unexplored issue

and examines how observing a robot expressing approach-avoidance conflict influences observers' self-evaluations of personal courage.

Design of the Video Stimuli

Robot and Projection Environment

The robot used in this study had the same body design as the switch-based robot interface proposed by Omichi et al. (2026). This robot has a cylindrical body; when it is not speaking, an agent with a face is retracted inside the body, and when it speaks, the agent emerges from the top of the body (Omichi et al., 2026). Figure 1 shows the robot used in both studies in its non-speaking and speaking states. As this study examined whether presenting the robot's internal state before action through speech bubbles influences observers' self-evaluations of personal courage, it was necessary to minimize interpretive variability arising from the robot's appearance or bodily movements. Unlike humanoid robots, this robot does not have complex facial expressions or limb movements; its expressive modalities are primarily the displayed face and speech. This design made it suitable as a stimulus medium for presenting approach motives, avoidance motives, and the presence or absence of admonishing speech in a relatively controlled manner. The admonishing speech was prerecorded and triggered by the experimenter using keyboard input.

In the video stimuli, the robot was filmed such that it appeared in the lower-right corner of the frame, and the robot's internal state was displayed as a speech bubble projected onto the wall behind the robot. The speech bubble was positioned near the robot's head and was used as a visual expression of the internal state before action rather than as the robot's spoken output.

Stimulus Scenario

In this study, we used video stimuli depicting a scene in which a robot observes a person littering in a park. Littering in a park is readily understood as a violation of shared norms in a public space. Indeed, littering has been used as a representative task for examining the effects of descriptive and injunctive norms (Cialdini et al., 1990). Observational research in public spaces has also shown that the presence of existing litter increases littering, whereas the availability of trash bins reduces it. Thus, littering can be positioned as an everyday norm violation affected by environmental and normative cues (Schultz et al., 2013). In addition, cues of disorder, such as litter and graffiti, can induce other norm violations (Keizer et al., 2008). Therefore, admonishing the person in this scene has value because it may help keep the park clean, create an opportunity for the person to stop littering, and reaffirm the norm that littering is unacceptable. At the same time, admonishing a stranger involves social and psychological risks, such as being yelled at, ignored, or viewed negatively. Littering in a park has also been used as a stimulus scenario in prior research on social control behavior, supporting its validity as a scenario involving intervention in a stranger's norm violation (Chekroun and Brauer, 2002). Field research has further shown that sanctions against strangers' norm violations include not only direct admonition but also indirect methods, and direct admonition can involve costs, including possible retaliation (Balafoutas et al., 2014). Research on social courage has identified moving toward valued action while facing interpersonal risk as an important feature (Howard et al., 2017). Therefore, admonishing a person for littering in a park is suitable as a stimulus scenario for examining personal courage because it involves both valued action and interpersonal risk and can elicit hesitation or conflict for the actor, although it does not involve extreme physical danger.

An example of the scenario description in the video stimulus is shown in Figure 2.

Common Structure of the Video Stimuli

The video stimuli did not present the park scene itself as live-action footage; instead, the scene was constructed using a textual scenario description and speech-bubble expressions. At the beginning of the video, the following scenario description was presented: "The robot saw a person littering in a park." The robot then displayed situation recognition in the speech bubble, such as "Oh, there is a person littering," followed by a statement indicating that admonition was required. The subsequent presentation of motives and the presence or absence of admonishing speech were manipulated according to the conditions in each study. All videos were created at a resolution of 1280 × 720 pixels. The duration of each segment was set to allow participants sufficient time to read the on-screen text at a natural pace. Minor differences in total duration arose from differences in the wording and presentation format of the condition-specific motive statements. Consequently, the videos lasted approximately 50–54 s in Study 1 and 49–51 s in Study 2.

Approach Motives and Avoidance Motives

In this study, reasons for moving toward a valued action were defined as approach motives, and reasons for avoiding the risks or disadvantages associated with the action were defined as avoidance motives. The specific motive statements are shown in Table 1. The approach motives indicated the possibility that desirable outcomes would result from admonishing behavior, whereas the avoidance motives indicated social and psychological risks that could result from admonishing behavior. By combining these approach and avoidance motives, we expressed a state in which reasons for moving toward a valued action and reasons for avoiding that action coexist. We operationalized this state as approach-avoidance conflict, in which approach and avoidance motives operate simultaneously (Lewin, 1931; Miller, 1944).

An example in which the approach and avoidance motives were presented simultaneously is shown in Figure 3.

By applying this common design, internal states before action are presented herein in a form that observers could visually interpret. The details of the condition manipulations in each study are described in the respective method sections.

Study 1: Is a Robot Expressing Approach-Avoidance Conflict Perceived as Courageous?

Purpose

In Study 1, we examined whether the presentation intended for use in Study 2 would be perceived as courageous. Specifically, we examined whether a robot expressing approach-avoidance conflict as an internal state would be perceived as more courageous than a robot not expressing conflict. We also used conflict ratings to confirm whether the presented internal state was perceived as approach-avoidance conflict and conducted an exploratory comparison of the effects of sequential and simultaneous presentation on courage and conflict ratings.

Approach-avoidance conflict is described as a state in which approach and avoidance motives compete with respect to the same goal (Lewin, 1931; Miller, 1944; Gray and McNaughton, 2000). Based on this view, simultaneous presentation was considered a candidate method for representing the coexistence of both motives at the same point in time. Ambivalent affect, meanwhile, has been argued to arise not only from the simultaneous co-occurrence of positive and negative affect but also from rapid alternation between them (Vaccaro et al., 2020). Although ambivalent affect and approach-avoidance motives are not identical

constructs, this account informed the presentation design. In the present study, however, the switching speed was reduced so that observers could visually identify each motive. Sequential presentation therefore represented temporal alternation rather than rapid alternation per se. Because these prior studies did not directly compare presentation methods for robots, the two methods were compared exploratorily in Study 1.

Based on this reasoning, we predicted that the conflict condition would yield higher robot courage ratings than the no-conflict condition. As a manipulation check, we also predicted that the conflict condition would yield higher conflict ratings than the no-conflict condition. The comparison between sequential and simultaneous presentation was exploratory and was conducted to examine which method would be perceived as more courageous.

Method

Conditions

Study 1 used a within-participant design with conflict and presentation method as factors. Conflict had two levels: no conflict and conflict. Presentation method also had two levels: sequential presentation and simultaneous presentation. To avoid confounding the effects of conflict and presentation method with the presence or absence of final action, all Study 1 conditions showed the robot performing the admonishing action.

In the stimulus videos, after the robot observed a person littering in a park, it displayed situation recognition and motives through a speech bubble. In the conflict condition, approach and avoidance motives were presented; in the no-conflict condition, only approach motives were presented. In the simultaneous presentation condition, the relevant motives were presented simultaneously and alternately enlarged and reduced. In the sequential presentation condition, the relevant motives were presented in sequence. Finally, in all conditions, the agent with a face emerged from the top of the body and said, "Excuse me, this is not a place to throw away trash." The four stimulus conditions used in Study 1 are summarized in Table 2.

Representative frames illustrating the presentation methods in the conflict conditions of Study 1 are shown in Figure 4. In Figure 4, the scenario description, situation recognition, and statement indicating that the situation required admonition are omitted; only the portions presenting the approach and avoidance motives are shown. In the sequential presentation condition, the avoidance motive "I would hate it if they yelled at me after I warned them" and the approach motive "If I warn them, the park might become cleaner" were presented in separate frames in sequence. In the simultaneous presentation condition, the same avoidance and approach motives were presented simultaneously within a single speech bubble and alternately emphasized.

Measurement

The dependent variables were robot courage ratings and conflict ratings. Each rating was obtained on a 7-point scale after participants observed each video, and the mean of the items was used as the scale score. Courage ratings were created by modifying the six items of the Japanese version of the Courage Measure (CM-J) to rate the robot (Shimotsukasa et al., 2023). The CM-J measures the tendency to act despite fear or anxiety, and its reliability and validity have been confirmed (Shimotsukasa et al., 2023). In Study 1, to measure the extent to which the robot appeared courageous rather than the observer's own courage, the subject of each item was changed to "This robot," and the item endings were changed to observational rating

expressions like “appeared to” and “was.” In the analysis, the mean of the six items shown in Table 3 was used as the courage rating score.

Conflict ratings were measured using study-specific items to assess the extent to which the robot appeared to waver between approach and avoidance motives. Based on the definition of approach-avoidance conflict (Lewin, 1931; Miller, 1944), we formulated items to capture observable aspects of this conflict: wanting to act but hesitating because of fear, showing hesitation or fluctuation in action, having mixed feelings about acting, and being unable to decide on an action. In the analysis, the mean of the four items shown in Table 4 was used as the conflict rating score. In addition, to confirm whether ratings based on these four items corresponded to a direct judgment of conflict, we also included a direct item: “This robot appeared to be experiencing conflict.”

Procedure

Study 1 was conducted via Yahoo! Crowdsourcing. Participants viewed videos from the four conditions, and after each video, completed the robot courage and conflict rating items. The order of video presentation was randomized for each participant. The questionnaire included an item asking about prior participation in the same survey, video-content comprehension checks, and attention-check items.

Participants

To assess the adequacy of the final sample size, we conducted a sample-size calculation in G*Power 3.1.9.7 (Faul et al., 2009) for the main effect of conflict, which tested the primary hypothesis. Because this main effect was a two-level comparison between the conflict and no-conflict conditions averaged across presentation methods, we selected “ANOVA: Repeated measures, within factors” under F tests and specified one group and two measurements. The “as in Cohen (1988) – recommended” effect-size specification was used, with $f(V) = 0.25$, $\alpha = 0.05$, power $(1 - \beta) = 0.80$, and nonsphericity correction $\epsilon = 1$. The resulting required sample size was 128 participants.

A total of 211 responses were recorded. Of these, 26 with missing responses to the check items were excluded. Among the remaining 185 responses, 54 met at least one of the following exclusion criteria: prior participation in the same survey ($n = 20$), failure to answer the attention-check item correctly ($n = 29$), or failure to answer at least one of the four video-content comprehension items correctly ($n = 28$); these criteria overlapped. Thus, 80 responses were excluded in total, yielding a final analytic sample of 131 participants.

Participants were 18–29 years old, with a mean age of 24.26 years ($SD = 3.36$). The sample included 76 women, 52 men, and 3 participants who selected “do not know/prefer not to answer.”

Statistical Analysis

In Study 1, robot courage rating scores and conflict rating scores were treated as dependent variables. Two-way repeated-measures analyses of variance were conducted with conflict (no conflict vs. conflict) and presentation method (sequential vs. simultaneous) as within-participant factors. Partial η^2 was reported as the effect size. For the courage ratings and conflict ratings, internal consistency was examined using Cronbach’s α , and the validity of treating the mean of each item set as a scale score was further examined using factor analysis with a one-factor solution. When an interaction was significant, the normality of the difference scores for paired comparisons between conditions was examined using the Shapiro–Wilk test; when normality was

rejected, Wilcoxon signed-rank tests were used. Analyses were conducted using the Python packages pandas, pingouin, and SciPy.

Results

First, we examined whether the courage ratings and conflict ratings could each be treated as a unidimensional scale score. Because the sign of factor loadings obtained in factor analysis is arbitrary, factor loadings are reported as absolute values. Cronbach's α for the six courage-rating items was 0.925. In the factor analysis with a one-factor solution, factor loadings ranged from 0.727 to 0.855, and the variance explained was 68.0%. The first eigenvalue was 4.396, and all subsequent eigenvalues were below 1. Therefore, the mean of the six items was used as the courage rating score. Cronbach's α for the four conflict-rating items was 0.928. In the factor analysis with a one-factor solution, factor loadings ranged from 0.845 to 0.898, and the variance explained was 76.5%. The first eigenvalue was 3.293, and all subsequent eigenvalues were below 1. Furthermore, a high positive correlation was found between the factor score calculated from the four items and the direct item ($r = 0.887$, $p < 0.001$). Therefore, the mean of the four items was used as the conflict rating score.

For courage ratings, the conflict condition was rated higher than the no-conflict condition. The main effect of conflict was significant ($F(1, 130) = 12.216$, $p < 0.001$, partial $\eta^2 = 0.086$). In contrast, the main effect of presentation method was not significant ($F(1, 130) = 2.657$, $p = 0.106$, partial $\eta^2 = 0.020$), and the interaction was also not significant ($F(1, 130) = 0.906$, $p = 0.343$, partial $\eta^2 = 0.007$).

Condition means are shown in Figure 5.

For conflict ratings, the conflict condition was rated higher than the no-conflict condition, and simultaneous presentation was rated higher than sequential presentation. The main effect of conflict was significant ($F(1, 130) = 79.558$, $p < 0.001$, partial $\eta^2 = 0.380$). The main effect of presentation method was also significant ($F(1, 130) = 46.448$, $p < 0.001$, partial $\eta^2 = 0.263$), and the interaction was significant ($F(1, 130) = 4.924$, $p = 0.028$, partial $\eta^2 = 0.036$).

Because the interaction was significant, paired comparisons between conditions were conducted. Shapiro-Wilk tests indicated significant departures from normality for all difference scores (all $p < 0.001$); therefore, Wilcoxon signed-rank tests were used. Under sequential presentation, conflict ratings were higher in the conflict condition ($M = 4.964$) than in the no-conflict condition ($M = 3.615$; $W = 671.5$, $p < 0.001$). The same pattern was observed under simultaneous presentation (conflict $M = 5.344$, no-conflict $M = 4.303$; $W = 882.5$, $p < 0.001$). Simultaneous presentation also yielded higher ratings than sequential presentation in both the no-conflict condition ($W = 1069.0$, $p < 0.001$) and the conflict condition ($W = 1467.0$, $p < 0.001$).

Condition means are shown in Figure 6.

Discussion

In Study 1, the robot expressing approach-avoidance conflict was rated as more courageous than the robot in the no-conflict condition. This indicates that even when the overt behavior was the same, showing conflict between approach and avoidance motives before action made the behavior more likely to be interpreted as involving fear or difficulty for the actor. This interpretation is consistent with the view that actions involving high personal courage are accompanied by fear or difficulty (Pury et al., 2007). Therefore, externalizing internal

states related to personal courage before action may be effective for presenting the robot's behavior as courageous to observers.

Conflict ratings were also higher in the conflict condition than in the no-conflict condition. This indicates that the presentation of approach and avoidance motives through speech bubbles was perceived by observers as approach-avoidance conflict. Regarding presentation method, no clear difference in courage ratings was found between sequential and simultaneous presentation. By contrast, conflict ratings were higher under simultaneous than sequential presentation in both the conflict and no-conflict conditions. Thus, simultaneous presentation may have made the robot appear generally more conflicted, but the results do not indicate that it strengthened the distinction between the conflict and no-conflict conditions.

In Study 1, we confirmed that a robot expressing approach-avoidance conflict is perceived as courageous and provided evidence supporting the validity of the presentation to be used in Study 2. On this basis, Study 2 used simultaneous presentation, which yielded the highest mean conflict rating within the conflict condition, to examine how observing the robot was associated with observers' self-evaluations of personal courage.

Study 2: Does Observing a Robot Perceived as Courageous Influence Observers' Self-Evaluations of Personal Courage?

Purpose

In Study 2, we examined how observing the robot presentation shown in Study 1 to be perceived as courageous influenced observers' self-evaluations of personal courage. We also examined whether this influence differed depending on observers' preexisting courage tendency.

Study 2 crossed motivational structure (univalent vs. conflicted) with behavioral outcome (admonition vs. no admonition). This design reflected the view that courage involves both internal difficulty and movement toward valued action (Rachman, 1984; Woodard and Pury, 2007; Norton and Weiss, 2009). Courage has also been described as a sequential process involving situation perception, value evaluation, evaluation of action feasibility, and action decision (Chowkase et al., 2024). In the univalent conditions, motive direction was congruent with the final behavior: approach motives preceded admonition, whereas avoidance motives preceded non-action. In the conflicted conditions, both approach and avoidance motives were presented regardless of the final behavior. The resulting conditions allowed us to compare univalent and conflicted motivational structures within each behavioral outcome and to examine the added role of action when the robot expressed conflict.

In Study 2, we predicted that the effects of approach-avoidance conflict and action on observers' self-evaluations of personal courage would differ depending on their preexisting courage tendency. In particular, we predicted that, among participants with a low preexisting courage tendency, personal courage self-evaluation scores after observation would be highest in the conflict-with-action condition. The rationale was that the low-courage group, whose self-evaluations of courage were low, was expected to experience difficulty acting in conflict situations. Prior research has shown that observing a model who gradually copes while showing difficulty can increase self-efficacy and task performance (Schunk et al., 1987; Schunk and Hanson, 1989). It has also been shown that when a task is difficult, people are more likely to use others' judgments and behavior as cues, whereas people with high self-efficacy are less susceptible to social influence (Lucas et al., 2006). The conflict-with-action condition was designed to present both the coexistence of approach and avoidance motives and the choice to move toward action. In other words, it presents a process similar to the difficulty faced by the low-courage group while showing that this difficulty can be overcome.

Therefore, we predicted that personal courage self-evaluation scores after observation would be highest in the low-courage group in this condition.

Method

Conditions

In Study 2, we used a three-factor mixed design with preexisting courage tendency group as a between-participant factor and conflict and action as within-participant factors. Participants were classified into low-courage and high-courage groups based on their pre-stimulus CM-J scores. The no-conflict conditions presented a single motive direction congruent with the robot's final behavior, whereas the conflict conditions presented approach and avoidance motives simultaneously. Action was manipulated according to whether the agent with a face emerged from the top of the body and delivered the admonishing speech.

As in Study 1, the stimulus videos presented a scene in which the robot observed a person littering in a park. After displaying situation recognition, the robot presented the relevant motive statements in the speech bubble. In the conflict conditions, approach and avoidance motives were presented simultaneously and alternately enlarged and reduced. In the no-conflict conditions, motive content was congruent with the final behavior: only approach motives were presented in the action-present condition, whereas only avoidance motives were presented in the action-absent condition.

In the action-present condition, after the motive presentation, the agent emerged from the top of the body and said, "Excuse me, this is not a place to throw away trash." In the action-absent condition, after the motive presentation, the agent remained retracted inside the body and did not speak. The four stimulus conditions used in Study 2 are summarized in Table 5.

Measurement

In Study 2, we measured pre-stimulus personal courage tendency, post-stimulus self-evaluations of observers' own personal courage, and robot conflict ratings. Responses to the CM-J and conflict-rating items used in the analyses were provided on a 7-point scale. Personal courage was measured using the six items of the Japanese version of the Courage Measure (CM-J) (Shimotsukasa et al., 2023). The CM-J measures trait-level individual differences in courage to act despite fear or anxiety, and its reliability and validity have been confirmed (Shimotsukasa et al., 2023). In Study 2, we used the pre-stimulus CM-J score as an index of observers' preexisting courage tendency. By contrast, the post-stimulus CM-J score was treated not as a change in long-term courage trait itself but as the observer's self-evaluation of personal courage immediately after stimulus presentation. In the analysis, the mean of the six items was used as the personal courage self-evaluation score.

Conflict ratings were measured using the same four items as in Study 1. In the analysis, the mean of the four items was used as the conflict rating score. In addition, to confirm whether conflict ratings based on these four items corresponded to a direct judgment of conflict, the same direct item used in Study 1, "This robot appeared to be experiencing conflict," was also included.

Procedure

Study 2 was conducted via Yahoo! Crowdsourcing using SurveyMonkey. Participants first completed the CM-J before stimulus presentation. The four stimulus-response blocks were then presented in a randomized order for each participant. Each block contained one video followed by the self-evaluation items for the participant's

own personal courage and the robot conflict-rating items. The questionnaire also included an item asking about prior participation in the same survey, video-content comprehension checks, and attention-check items.

Participants

To assess the adequacy of the sample size, we conducted a power analysis in G*Power 3.1.9.7 (Faul et al., 2009) for the primary test: the three-way interaction among preexisting courage tendency group, conflict, and action. This interaction was specified as a one-degree-of-freedom effect comparing the within-participant conflict $\times$ action contrast between the low- and high-courage groups. Using “ANOVA: Fixed effects, special, main effects and interactions” under F tests, with $f = 0.25$, $\alpha = 0.05$, power $(1 - \beta) = 0.80$, numerator $df = 1$, and two groups, the required sample size was 128 participants.

Anticipating missing data and exclusions, we recruited 212 participants. Of these, 13 did not complete the survey and were excluded. Among the remaining 199 participants, 73 met at least one of the following exclusion criteria: prior participation in the same survey ($n = 19$), failure to answer the attention-check item correctly ($n = 31$), or failure to answer at least one of the four video-content comprehension items correctly ($n = 53$); these criteria overlapped. Thus, 86 participants were excluded in total, yielding a final analytic sample of 126. Although the final sample was two participants below the required sample size, the achieved power based on the same assumed effect size and the actual group sizes was 0.792, which was close to the planned value of 0.80.

Participants were 18–29 years old, with a mean age of 24.11 years ($SD = 3.62$). The sample included 55 men and 71 women. Based on pre-stimulus CM-J scores, participants with scores below 4 were classified as the low-courage group, and participants with scores of 4 or higher were classified as the high-courage group. The cutoff of 4 corresponded to the midpoint of the 7-point scale and distinguished scores below the scale midpoint from those at or above it. The low-courage group included 69 participants, and the high-courage group included 57 participants.

Statistical Analysis

In Study 2, post-stimulus personal courage self-evaluation scores were analyzed as the main outcome, and conflict rating scores were analyzed as a manipulation check. We conducted a three-way mixed analysis of variance with preexisting courage tendency group, based on pre-stimulus CM-J scores (below 4 vs. 4 or higher), as a between-participant factor and conflict (no conflict vs. conflict) and action (no action vs. action) as within-participant factors. Partial η^2 was reported as the effect size. When interactions were significant, simple effects were examined as needed. For simple-effects tests, the normality of paired differences was examined using the Shapiro–Wilk test; paired t-tests were used when normality was satisfied, and Wilcoxon signed-rank tests were used when normality was not satisfied. Conflict rating scores were analyzed using the same factor structure to confirm the success of the conflict manipulation. Analyses were conducted using the Python packages pandas, statsmodels, and SciPy.

Results

A three-way mixed analysis of variance on observers' personal courage self-evaluation scores revealed a significant main effect of preexisting courage tendency group ($F(1, 124) = 124.884$, $p < 0.001$, partial $\eta^2 = 0.502$). By contrast, neither the main effect of conflict ($F(1, 124) = 0.067$, $p = 0.796$, partial $\eta^2 < 0.001$) nor the main effect of action ($F(1, 124) = 2.487$, $p = 0.117$, partial $\eta^2 = 0.020$) was significant. Among the two-way

interactions, the interaction between preexisting courage tendency group and conflict was significant ($F(1, 124) = 7.513, p = 0.007, \text{partial } \eta^2 = 0.057$). By contrast, neither the interaction between preexisting courage tendency group and action ($F(1, 124) = 0.009, p = 0.925, \text{partial } \eta^2 < 0.001$) nor the interaction between conflict and action ($F(1, 124) = 2.613, p = 0.109, \text{partial } \eta^2 = 0.021$) was significant. The primary three-way interaction was also not significant ($F(1, 124) = 0.046, p = 0.831, \text{partial } \eta^2 < 0.001$). Thus, the hypothesis that personal courage self-evaluation scores would be particularly high in the conflict-with-action condition among the low-courage group was not supported.

Because the interaction between preexisting courage tendency group and conflict was significant, we examined the simple effect of conflict within each group. Shapiro–Wilk tests of the paired difference scores indicated no significant departure from normality in the low-courage group ($W = 0.974, p = 0.152$) but indicated a significant departure in the high-courage group ($W = 0.928, p = 0.002$). Accordingly, a paired t -test was used for the low-courage group, whereas a Wilcoxon signed-rank test was used for the high-courage group. The simple-effects analyses showed a marginal trend toward higher personal courage self-evaluation scores in the conflict condition in the low-courage group (conflict $M = 3.104$, no-conflict $M = 2.982, t(68) = 1.980, p = 0.052, d = 0.238$). In contrast, participants in the high-courage group reported significantly lower personal courage self-evaluation scores in the conflict condition (conflict $M = 4.757$, no-conflict $M = 4.858, W = 404.0, p = 0.038, d = -0.271$).

These simple effects are shown in Figure 7.

As a manipulation check, conflict ratings were higher in the conflict condition than in the no-conflict condition. The main effect of conflict was significant ($F(1, 124) = 52.939, p < 0.001, \text{partial } \eta^2 = 0.299$). A supplemental check also showed that the conflict condition was rated higher than the no-conflict condition ($W = 1128.0, p < 0.001, \text{conflict } M = 5.151, \text{no-conflict } M = 4.209$). Therefore, the conflict manipulation was successful in Study 2 as well.

Discussion

In Study 2, the hypothesis that the low-courage group would show the highest personal courage self-evaluation scores in the conflict-with-action condition was not supported. This hypothesis treated the robot that expressed conflict and then admonished the litterer as analogous to a coping model that displays difficulty but ultimately performs the task. However, neither the main effect of action nor any interaction involving action was significant. The relative advantage of coping models has also not been consistently observed in prior research: Schunk and Hanson (1985) found no significant differences in self-efficacy or achievement between mastery- and coping-model conditions. Accordingly, the failure to detect a specific advantage for the conflict-with-action condition is compatible with evidence that coping models do not invariably outperform mastery models.

However, the interaction between preexisting courage tendency group and approach-avoidance conflict was significant, indicating that the conflict contrast differed between groups. In the low-courage group, there was a marginal trend toward higher self-evaluation scores in the conflict condition. Previous studies have shown that observing a model who gradually copes while displaying difficulty can, in some circumstances, enhance self-efficacy and task performance more than observing a model who succeeds easily from the outset (Schunk et al., 1987; Schunk and Hanson, 1989). Other work has shown that people are more likely to use others' judgments and behavior as cues when a task is difficult, whereas those with high self-efficacy are less susceptible to such influence (Lucas et al., 2006). The conflict condition in the present study showed a process

in which the robot held an approach motive despite also having an avoidance motive. For the low-courage group, this may have served as a cue that a motive to move toward valued action can coexist with fear or hesitation. However, preexisting courage tendency and self-efficacy are not identical constructs, and we did not measure whether participants perceived the robot as a model coping with difficulty. Moreover, because this result was only marginal, this interpretation should remain tentative.

By contrast, participants in the high-courage group reported lower personal courage self-evaluation scores in the conflict condition. Chowkase et al. (2024) theorized that courageous decisions involve weighing potential valued outcomes against the perceived risks associated with action. In an approach-avoidance task, Bach et al. (2014) experimentally showed that greater potential threat was accompanied by greater passive avoidance and behavioral inhibition. In the present conflict condition, the robot simultaneously expressed an approach motive indicating the value of admonishing the litterer and avoidance motives indicating risks such as being yelled at or ignored. These avoidance motives may therefore have been interpreted as information about the risks of acting, leading the high-courage group to consider not only the value of the action but also its associated risks and to evaluate their own courage more cautiously. However, neither of these prior studies directly showed that observing another agent's conflict lowers self-evaluation specifically among individuals with high preexisting courage. They therefore provide only a possible general mechanism for interpreting the present result.

Although the main effect of conflict on personal courage self-evaluation was not significant, the interaction between preexisting courage tendency group and conflict was significant, and the direction of the conflict contrast differed between groups. The absence of a main effect should therefore not be interpreted as evidence that conflict expression was unrelated to self-evaluation in both groups. Moreover, conflict ratings were higher in the conflict condition than in the no-conflict condition, indicating that participants perceived the intended conflict expression. Thus, the unsupported hypothesis cannot be attributed solely to a failed conflict manipulation. Overall, robot-expressed conflict did not uniformly raise observers' self-evaluations; instead, its association with self-evaluation may differ in direction according to observers' preexisting courage tendency.

General Discussion

The studies of robot-based social modeling reviewed in this manuscript have examined how observable behaviors, social roles, and forms of encouragement affect observers (Xu, 2023; Higashino et al., 2023). In contrast, the present study used a robot as a social model that externalized approach and avoidance motives before action in a controlled manner. The manipulation check in Study 1 confirmed that presenting both motives in speech bubbles was perceived as approach-avoidance conflict. This extends the use of speech bubbles from representing attention or mental focus (Nitada et al., 2021) to representing conflicting motives. In Study 1, manipulating the expressed motives while holding the robot's appearance and final behavior constant also enabled comparisons of pre-action internal states that are difficult to observe and control directly in human models. Thus, the contribution of this study is not to establish the superiority of robots over other presentation media, but to demonstrate an experimental approach for human-robot interaction research in which internal states can be presented under controlled conditions.

Taken together, the two studies indicate that how a robot is perceived and how observing it affects users should be considered separately when designing robot expressions of internal states. In Study 1, conflict expression increased ratings of the robot's courage, whereas in Study 2, the association between conflict expression and observers' self-evaluations of personal courage differed according to preexisting courage tendency. Furthermore, simultaneous presentation made the robot appear more conflicted, but it increased

conflict ratings not only in the conflict condition but also in the no-conflict condition. Thus, strengthening the impression of conflict is not the same as making the difference between the conflict and no-conflict conditions clearer. Accordingly, when designing robot expressions of internal states, researchers should separately evaluate how strongly the state is perceived, whether the intended state can be distinguished from other states, and how the expression affects users. Because the same conflict expression may not produce the same outcome for all users, expressions that take users' prior characteristics into account warrant further investigation. However, because the result for the low-courage group was only marginal and actual behavior was not measured, the present study does not establish the effectiveness of personalized expressions.

Limitations

This study has several limitations. First, although we measured observers' self-evaluations of personal courage in this study, we did not measure whether observers actually moved toward valued action. The CM-J measures trait-level individual differences in courage to act despite fear or anxiety (Shimotsukasa et al., 2023). Therefore, the post-stimulus CM-J score in this study should be interpreted not as a change in long-term courage trait but as observers' self-evaluations of personal courage immediately after stimulus presentation.

Second, in this study, we did not directly measure the psychological process through which the presentation of approach-avoidance conflict operated in different directions for the low-courage and high-courage groups. We suggested that, for the low-courage group, the presentation may have served as a cue that "a motive to move toward valued action can coexist with fear or hesitation," whereas for the high-courage group, it may have made avoidance motives or the difficulty of action more salient. However, because we did not directly measure whether the robot was perceived as a model coping with difficulty, perceived similarity to the robot, self-efficacy, interpretation of avoidance motives, perceived action difficulty, or processing of risk information, these explanations remain interpretive.

Third, in this study, we used only video stimuli involving a robot and did not compare other presentation media, such as a physical robot, human model, avatar, animation, or text presentation. Therefore, it is not possible to determine whether the present results are attributable to the robot as a nonhuman agent, the video format, or the explicit presentation of approach-avoidance conflict through a speech bubble.

Fourth, the stimulus scenario in this study was limited to a robot admonishing a person littering in a park. This scenario has some validity as a stimulus for examining personal courage because valued action and interpersonal risk coexist. However, in this study, we examined a scene in which the robot directly admonished a stranger's norm violation. Personal courage may include diverse situations, such as speaking up in interpersonal contexts, seeking help, challenging learning tasks, social reintegration, and prosocial behavior (Pury et al., 2007; Vogel et al., 2007; Howard et al., 2017). Therefore, it remains unclear whether the present findings generalize to situations involving different types of risk or action value.

Future Work

Future research should proceed in two directions: basic and applied approaches. Basic research should examine the psychological process through which presenting approach-avoidance conflict affects observers' self-evaluations of personal courage. Specifically, future studies should measure the extent to which observers perceive the robot as a self-relevant model or as a model coping with difficulty, perceived similarity to the robot, self-efficacy, perceived threat and difficulty, and the added value of action information. Such studies should also clarify the conditions under which the interaction between preexisting courage tendency and approach-avoidance conflict emerges. Future studies of how internal states are displayed should manipulate

simultaneous versus sequential presentation, the degree of emphasis, and presentation timing, and should assess not only how strongly the internal state is perceived but also whether the intended state can be correctly distinguished from other states. In addition, using the same scenario and internal-state expression, a physically present robot, a robot presented in a video, a human model, an avatar or animation, and text-only presentation should be compared in future studies. This comparison would help clarify the conditions in which a nonhuman robot functions as a social model and the roles played by physical embodiment, co-presence, and movement.

Applied research should examine how the outcomes associated with robot-expressed conflict vary across users and situations. In contexts such as education, counseling or social reintegration, and everyday prosocial behavior, future studies should measure not only self-evaluation but also actual behavior, including speaking up, asking questions, seeking help, pointing out problems, and helping others, as well as sustained changes in such behavior. In addition, designs that present the same conflict expression to all users should be directly compared with designs that adjust the content or intensity of the expression according to user characteristics such as preexisting courage tendency, thereby testing the effectiveness of personalized expressions.

Conclusion

This study examined how a robot's externalization of pre-action approach-avoidance conflict was related to perceptions of the robot's courage and observers' self-evaluations of personal courage. In Study 1, the robot received higher courage ratings when it expressed conflict; in Study 2, the direction of the difference in self-evaluation between the conflict and no-conflict conditions varied according to observers' preexisting courage tendency. This study positions robots as social models capable of presenting difficult-to-observe pre-action internal states under controlled conditions and suggests that the same expression may not have uniform consequences across users. However, these findings do not demonstrate an effect unique to robots or a change in actual courageous behavior. Future work should compare robots with other presentation media and test the effects of personalized expressions on actual behavior.

Conflict of Interest

The authors declare that this study received funding from DAIKIN INDUSTRIES, LTD. The funder was not involved in the study design, data collection, analysis, interpretation of data, the writing of this article, or the decision to submit it for publication.

Ethics Statement

The study was approved by the Ethics Committee of the Graduate School of Engineering Science at The University of Osaka (R2-8-7). Before beginning the survey, participants were informed of the study purpose, the voluntary nature of participation, their right to discontinue participation at any time without disadvantage, and the anonymity of their responses. All participants provided electronic informed consent by confirming this information and proceeding to the survey.

Author Contributions

The Author Contributions section is mandatory for all articles, including articles by sole authors. If an appropriate statement is not provided on submission, a standard one will be inserted during the production process. The Author Contributions statement must describe the contributions of individual authors referred to

by their initials and, in doing so, all authors agree to be accountable for the content of the work. Please see [here](#) for full authorship criteria.

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Data Availability Statement

The datasets [GENERATED/ANALYZED] for this study can be found in the [NAME OF REPOSITORY] [LINK]. Please see the “Availability of data” section of [Materials and data policies in the Author guidelines](#) for more details.

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Figure Captions

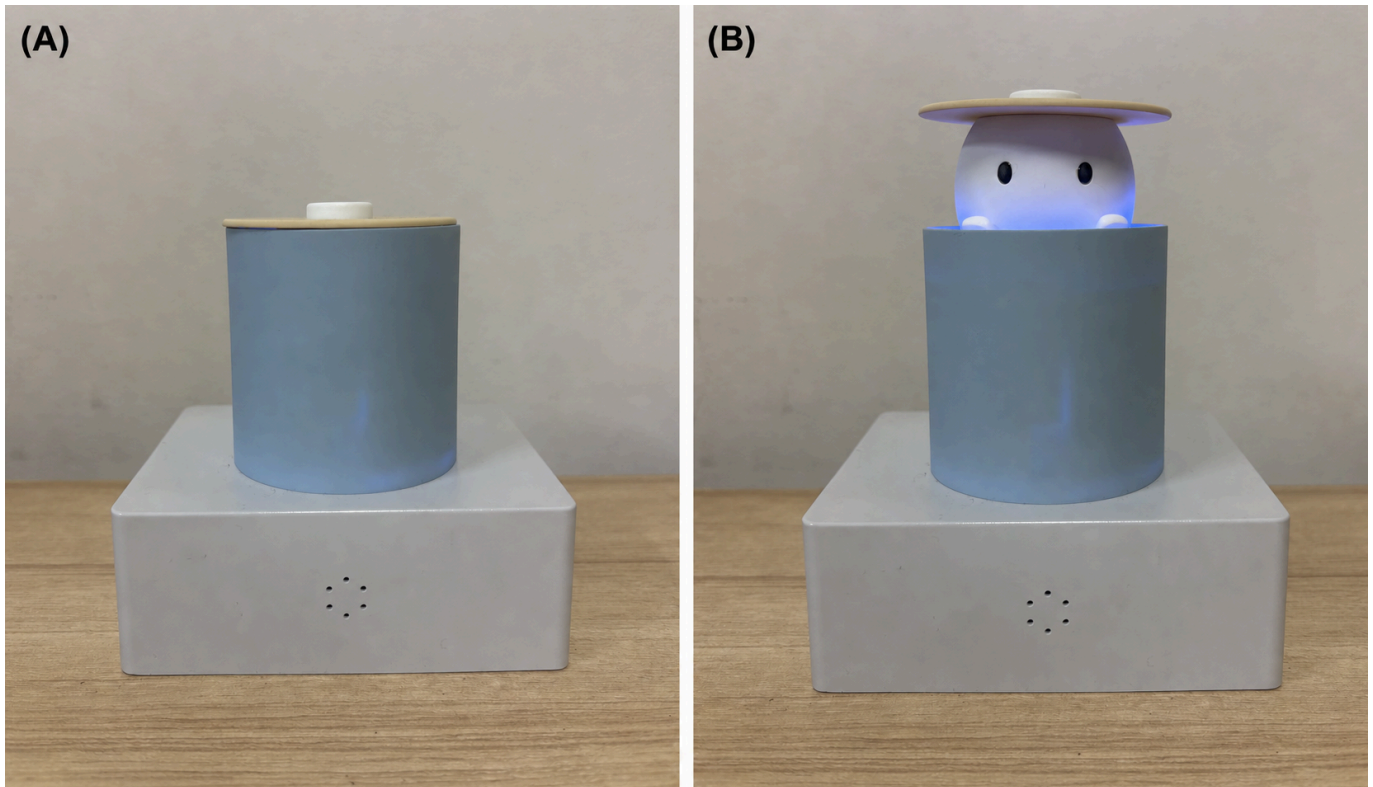


Figure 1. Appearance of the robot used in the video stimuli in both studies. (A) Non-speaking state, with the agent with a face retracted inside the cylindrical body. (B) Speaking state, with the agent extending above the body.

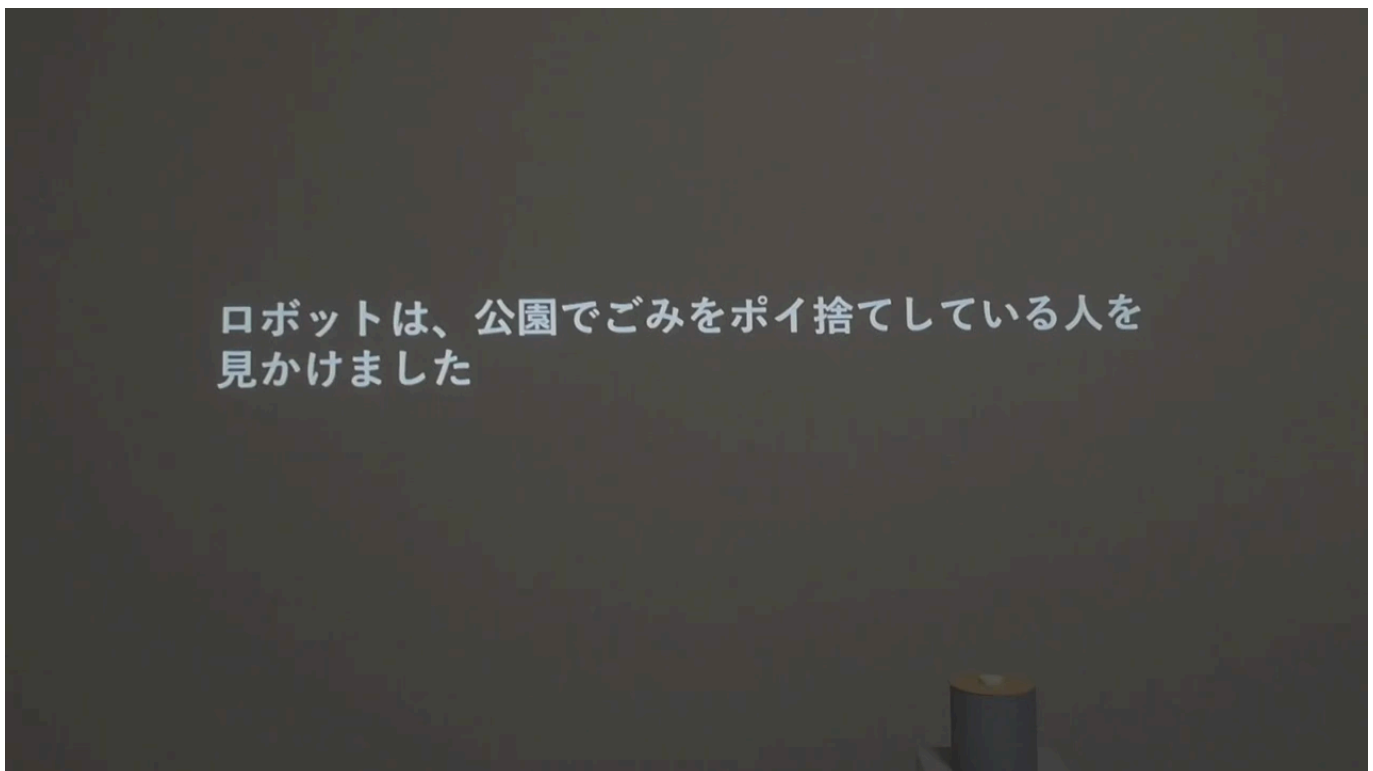


Figure 2. Example of the scenario description in the video stimulus. The text presents a situation in which the robot sees a person littering in a park.

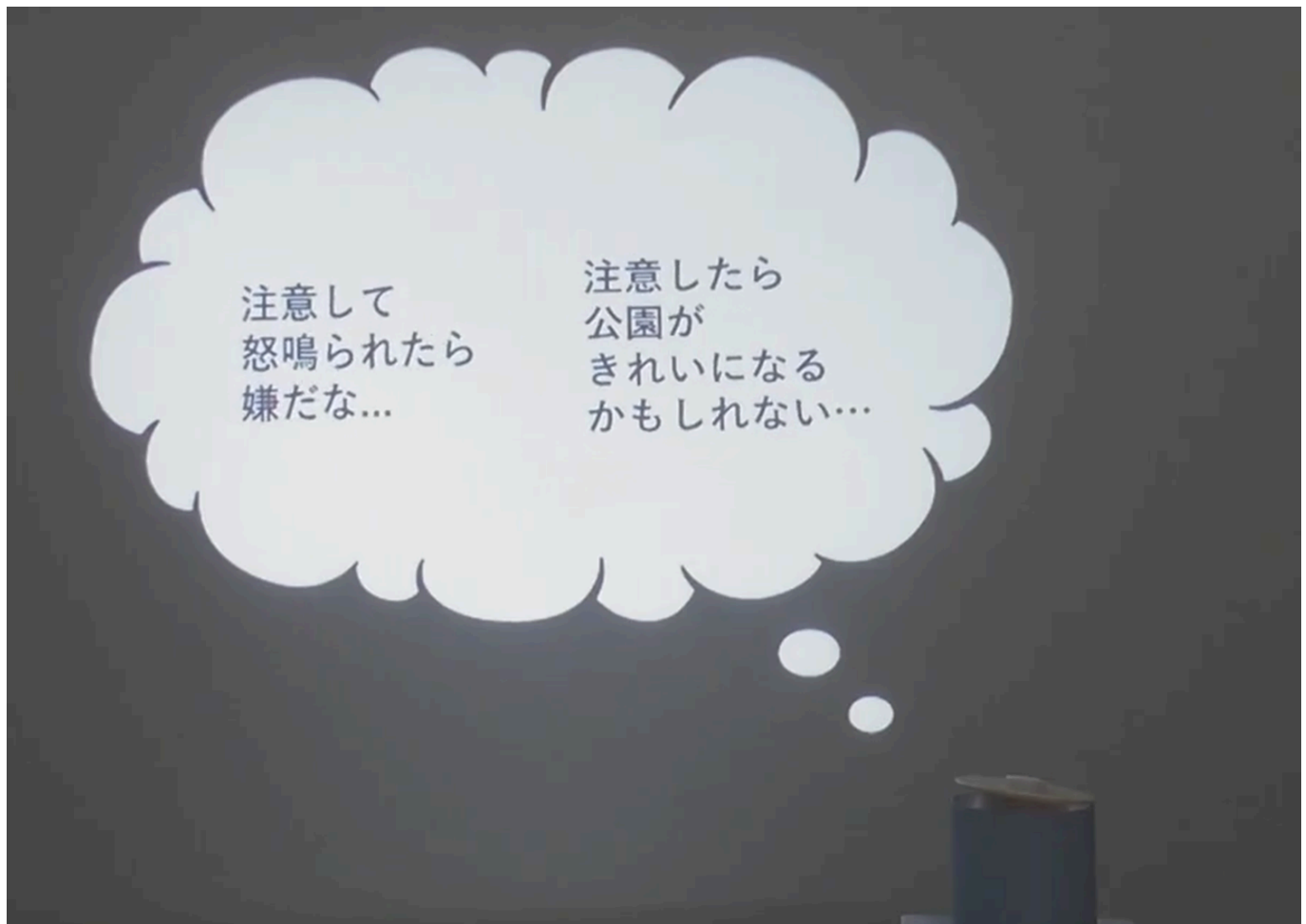
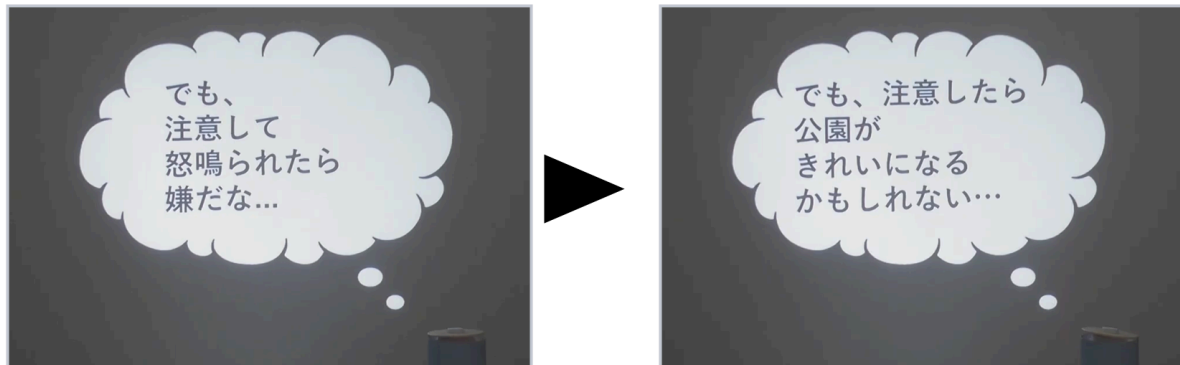


Figure 3. Example of simultaneous presentation of approach and avoidance motives. The speech bubble simultaneously displays a desirable consequence of admonition and a concern about admonition.

Study 1: Conflict Presentation

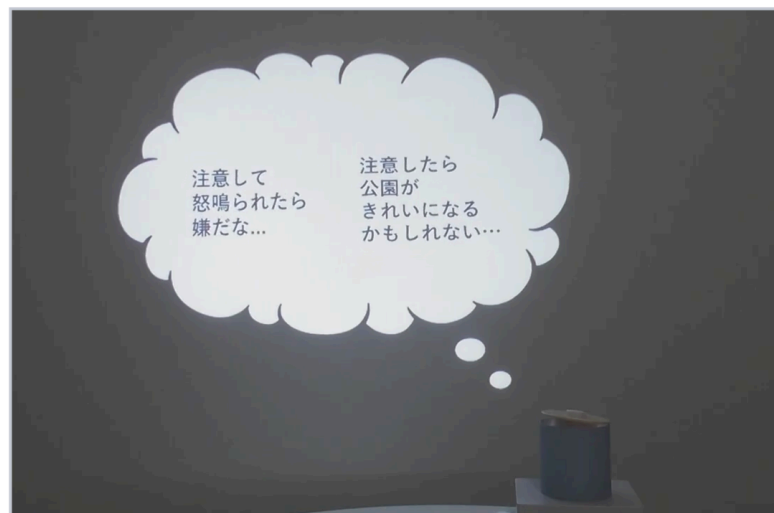
(A) Sequential: motives shown in separate frames



Avoidance motive
They might yell at me...

Approach motive
The park might become cleaner...

(B) Simultaneous: both motives shown in one frame



Both motives were displayed together

Figure 4. Comparison of conflict presentation methods in Study 1. (A) In the sequential presentation condition, the avoidance and approach motives were shown in separate frames. (B) In the simultaneous presentation condition, both motives were shown within a single frame.

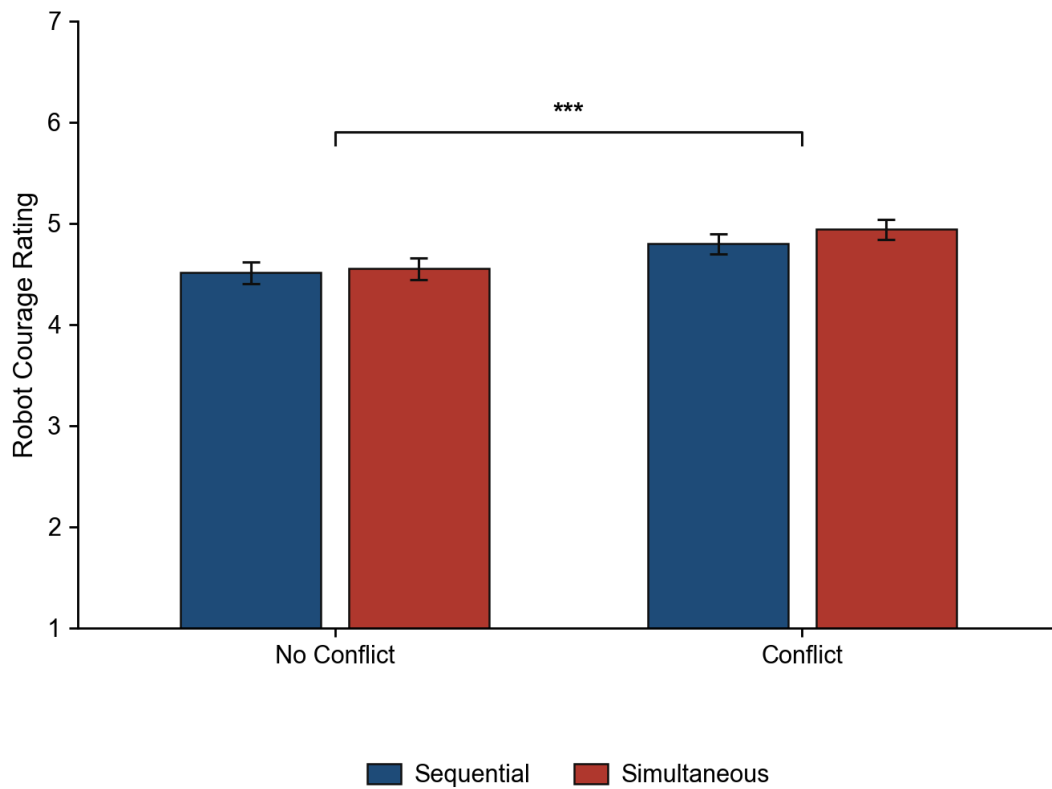


Figure 5. Condition means of courage ratings in Study 1.

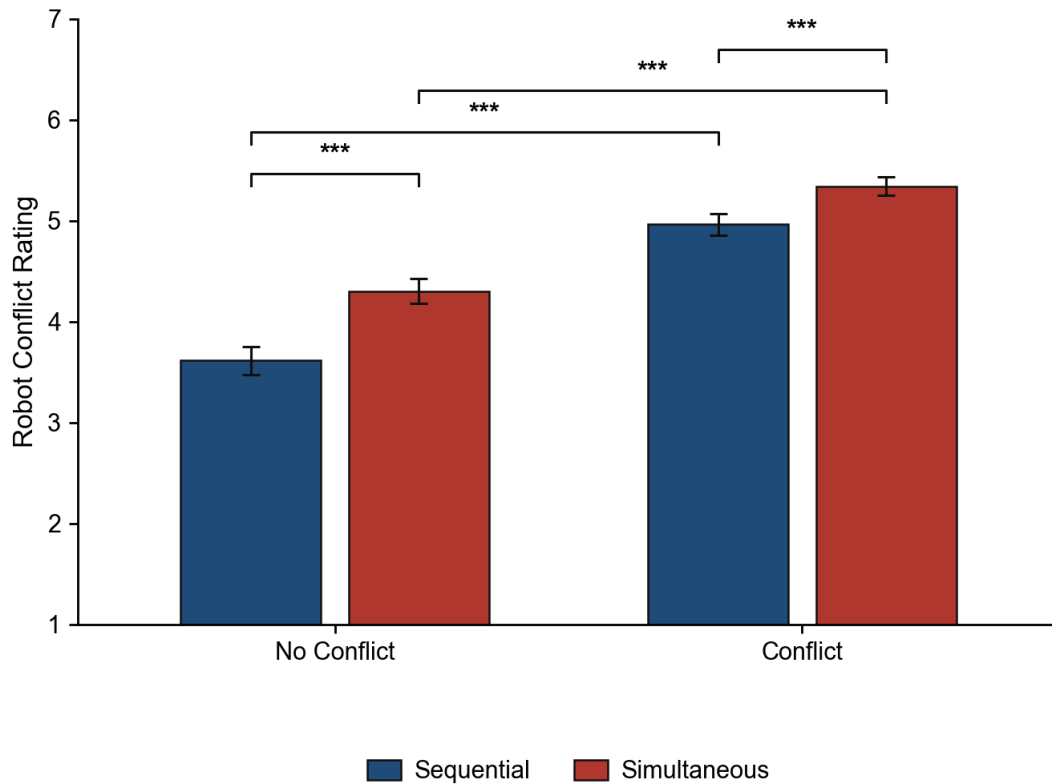


Figure 6. Condition means of conflict ratings in Study 1.

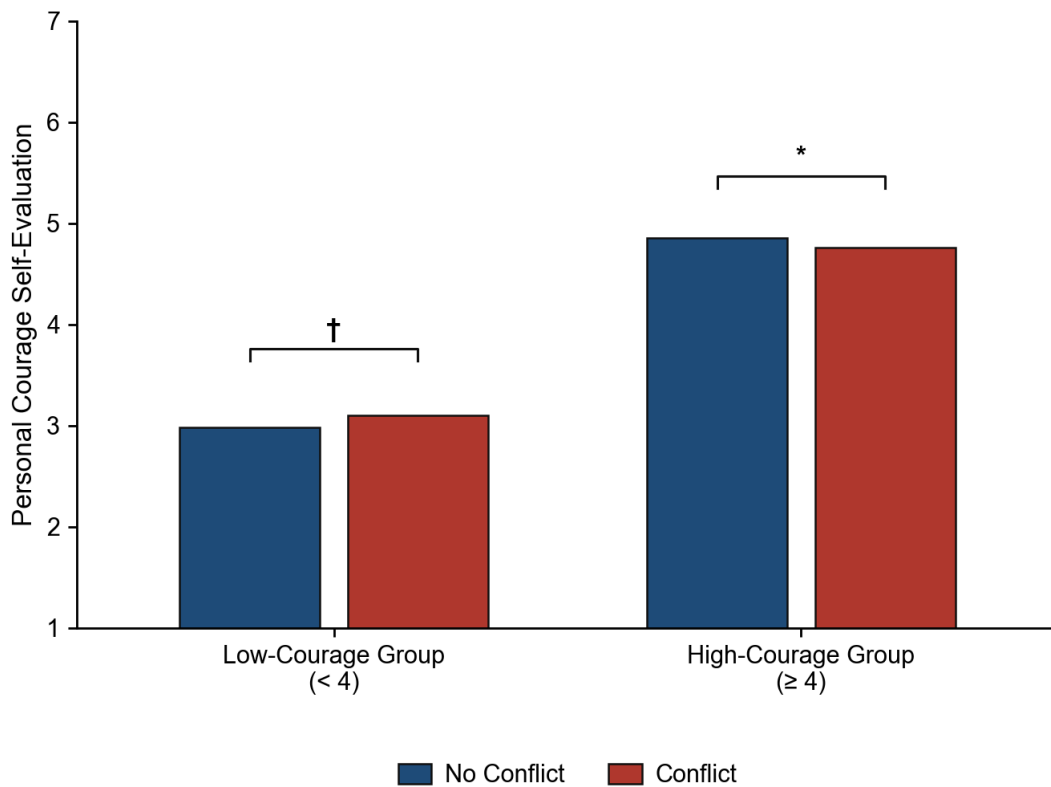


Figure 7. Simple effects of personal courage self-evaluations in Study 2. Means for the no-conflict and conflict conditions by preexisting courage tendency group. The dagger denotes a marginal trend ($p = 0.052$), and the asterisk denotes $p < 0.05$.

Tables

Table 1. Approach and avoidance motive statements used in the video stimuli.

Type of motive	Motive statement	Content
Approach motive	"If I warn them, the park might become cleaner."	Possibility that admonishing behavior improves the public space
Approach motive	"If I warn them, it might prompt that person to stop littering."	Possibility that admonishing behavior changes the other person's behavior
Avoidance motive	"I would hate it if they yelled at me after I warned them."	Risk of receiving an aggressive response from the other person because of admonishing behavior
Avoidance motive	"Even if I warn them, they might ignore me and not take me seriously."	Risk that the admonition is not accepted and is ignored

Table 2. Stimulus conditions in Study 1.

Study 1 condition	Internal-state content	Motive statements used	Presentation method	Final admonishing speech
No conflict, sequential presentation	Approach motives only	Approach motives: "If I warn them, the park might become cleaner"; "If I warn them, it might prompt that person to stop littering"	Approach motives presented in sequence	Present
No conflict, simultaneous presentation	Approach motives only	Approach motives: "If I warn them, the park might become cleaner"; "If I warn them, it might prompt that person to stop littering"	Multiple approach motives presented simultaneously and alternately emphasized	Present
Conflict, sequential presentation	Approach and avoidance motives	Approach motives: "If I warn them, the park might become cleaner"; "If I warn them, it might prompt that person to stop littering" / Avoidance motives: "I would hate it if they yelled at me after I warned them"; "Even if I warn them, they might ignore me and not take me seriously"	Approach and avoidance motives presented in sequence	Present
Conflict, simultaneous presentation	Approach and avoidance motives	Approach motives: "If I warn them, the park might become cleaner"; "If I warn them, it might prompt that person to stop littering" / Avoidance motives: "I would hate it if they yelled at me after I warned them"; "Even if I warn them, they might ignore me and not take me seriously"	Approach and avoidance motives presented simultaneously and alternately emphasized	Present

Table 3. Robot courage-rating items used in Study 1.

Item	Courage-rating item
1	This robot appeared to confront its own fear.
2	This robot appeared not to run away until it did what it had to do, even if it felt strong fear.
3	This robot did something even though it seemed dangerous.
4	This robot took action or confronted the situation anyway, even though it had some worry or anxiety.

Item	Courage-rating item
5	This robot confronted something frightening when there was an important reason to confront it.
6	This robot appeared not to back down, even when something threatened it.

Table 4. Robot conflict-rating items used in Studies 1 and 2.

Item	Conflict-rating item
1	This robot appeared to want to do what was in front of it but to hesitate because of fear.
2	This robot appeared to have hesitation or fluctuation in its own action.
3	This robot appeared to have mixed feelings of wanting and not wanting to act.
4	This robot appeared unable to decide its own action.

Table 5. Stimulus conditions in Study 2.

Study 2 condition	Internal-state content	Motive statements used	Presentation method	Final admonishing speech
No conflict, no action	Avoidance motives only	Avoidance motives: "I would hate it if they yelled at me after I warned them"; "Even if I warn them, they might ignore me and not take me seriously"	Avoidance motives presented and emphasized	Absent
No conflict, action	Approach motives only	Approach motives: "If I warn them, the park might become cleaner"; "If I warn them, it might prompt that person to stop littering"	Approach motives presented and emphasized	Present
Conflict, no action	Approach and avoidance motives	Approach motives: "If I warn them, the park might become cleaner"; "If I warn them, it might prompt that person to stop littering" / Avoidance motives: "I would hate it if they yelled at me after I warned them"; "Even if I warn them, they might ignore me and not take me seriously"	Simultaneous presentation with alternating emphasis	Absent

Study 2 condition	Internal- state content	Motive statements used	Presentation method	Final admonishing speech
Conflict, action	Approach and avoidance motives	Approach motives: "If I warn them, the park might become cleaner"; "If I warn them, it might prompt that person to stop littering" / Avoidance motives: "I would hate it if they yelled at me after I warned them"; "Even if I warn them, they might ignore me and not take me seriously"	Simultaneous presentation with alternating emphasis	Present